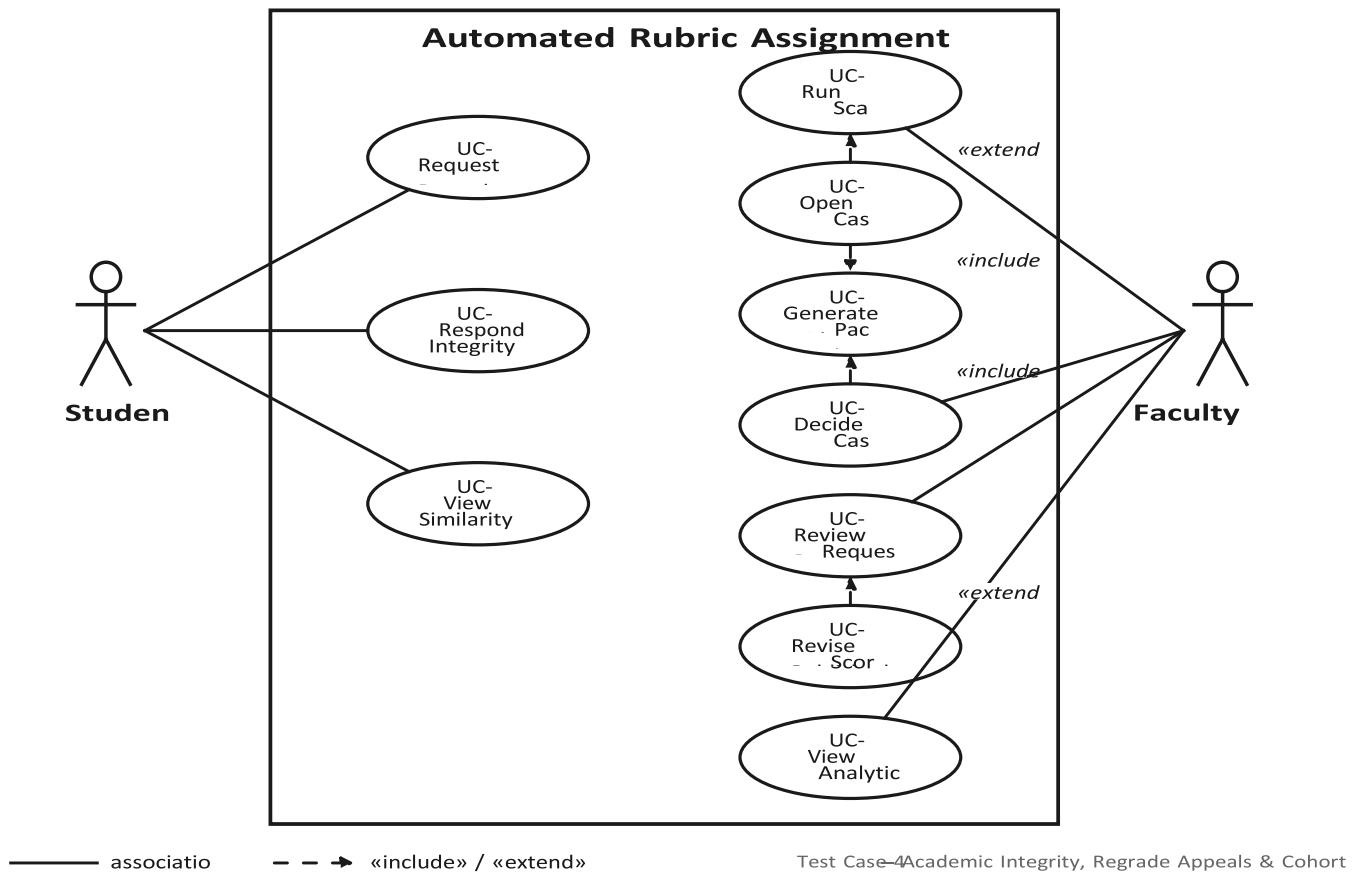


Requirements Table — Test Case 4: Academic Integrity, Regrade Appeals & Cohort Analytics

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional — Similarity Scan	The system shall run a token-level structural similarity scan of every submission against all other submissions in the cohort and against the archive of previous offerings, producing a similarity percentage and a matched-fragment report per submission pair.	High	Pass: a pair of submissions differing only in identifiers and formatting is reported above the high-similarity threshold with the matched fragments highlighted. Fail: a renamed-variable clone is scored as unrelated, or the report omits the matched fragments.	Structural comparison is the only method that survives cosmetic obfuscation.
FR-002	Functional — Integrity Case Workflow	The system shall let a Faculty Evaluator open an integrity case from a flagged pair, attach an automatically assembled evidence pack, notify the students, accept a written student response within a configured window, and record a decision with justification.	High	Pass: a case moves through opened → responded → decided with every transition, actor and timestamp recorded; a decision without justification is refused. Fail: a case is decided before the response window closes, or a transition is unlogged.	Integrity decisions are contestable and must follow due process with a complete record.
FR-003	Functional — Regrade Appeal	The system shall accept a regrade request from a Student within 72 hours of publication, naming the disputed criteria and a justification, route it to the Faculty Evaluator queue, and record the outcome with any revised score.	High	Pass: a request inside the window creates a tracked ticket visible to both parties; a request after the window is refused with the deadline shown. Fail: a late request is accepted, or a decided request leaves the published score inconsistent.	A bounded, tracked appeal path replaces ad-hoc email disputes and gives both sides a record.
FR-004	Functional — Score Revision	On an upheld appeal or integrity decision the system shall revise the published score, retain the previous value with the reason and deciding evaluator, notify the Student, and mark the submission as revised in all exports.	Medium	Pass: the revised score appears everywhere the original did, the original remains retrievable, and exports carry the revised marker. Fail: a revision overwrites history, or an export still shows the superseded score.	Grade corrections must be traceable end to end for institutional audit.
FR-005	Functional — Cohort Analytics & Export	The system shall present cohort analytics — score distribution, criterion-wise mean and pass rate, and a test-failure heat map — and shall export results, integrity outcomes and the audit log as CSV for the institutional record.	Medium	Pass: analytics recompute after any score revision, and the CSV export reconciles row for row with the on-screen figures. Fail: stale analytics after a revision, or an export that disagrees with the dashboard.	Criterion-wise failure data tells staff which topic to reteach and which test is mis-specified.
NFR-001	Privacy & Access Control	Similarity reports, integrity cases and cohort analytics shall be visible in identified form only to the Faculty Evaluator role; a Student shall see only their own report and case, and exports containing identities shall require an explicit elevated-role action over TLS 1.2+.	High	Pass: an access test confirms a student account cannot retrieve another student's similarity report, case or ranked position through any interface or API. Fail: any cross-student disclosure, or an unauthenticated export endpoint.	An unproven similarity flag is reputationally damaging and must never leak to peers.
NFR-002	Performance & Scalability	A full cross-cohort similarity scan of 500 submissions against the current cohort and two previous offerings shall complete within 15 minutes as a background job, and analytics queries shall return within 5 s.	Medium	Pass: the scan finishes inside the budget without blocking evaluation, and dashboards render within 5 s at full cohort size. Fail: the scan starves the evaluation queue, or analytics time out.	Pairwise comparison grows quadratically, so it must be scheduled off the interactive path.



Use-Case Flow Specification — Test Case 4

Use Case ID	UC-08
Use Case Name	Review Regrade Request
Primary Actor	Faculty Evaluator
Secondary Actors	Student (requester); notification service
Trigger	A Student submits a regrade request within the 72-hour window after publication
Related Requirements	FR-003, FR-004, FR-005, NFR-001
Included Use Cases	Extension: UC-09 Revise Published Score — applied only when the request is upheld

Preconditions

- The submission has been evaluated and its score published.
- The request was raised inside the 72-hour appeal window and names at least one disputed criterion.
- The evaluation report, execution logs and rubric version used are retrievable.
- The Faculty Evaluator is authenticated and owns the assignment.

Postconditions — success guarantee

- The request is marked decided with the outcome, justification, evaluator ID and timestamp.
- On an upheld request the published score is revised and the previous value is retained.
- The Student is notified with the decision and the reasoning.
- Cohort analytics and exports reflect the revised score.

Postconditions — minimal guarantee

- The request remains open and the published score is unchanged; no partial decision is recorded.

Main Success Scenario

1. The Faculty Evaluator opens the regrade queue and selects a pending request.
2. The system displays the disputed criteria beside the rubric version, test outcomes and execution logs used for the original score.
3. The Faculty Evaluator inspects the evidence for each disputed criterion.
4. The Faculty Evaluator records the outcome per criterion — upheld or rejected — with a justification.
5. The system recomputes the total from the revised criterion marks where the request is upheld — «*extend*» UC-09.
6. The system retains the previous score with the reason and marks the submission as revised.
7. The system closes the request, writes the decision to the audit log and notifies the Student.
8. The system recomputes cohort analytics and refreshes the export set.
9. The Student views the decision, the reasoning and the revised breakdown.

Alternate Flow A1 — request reveals a defect affecting the whole cohort

1. At step 3 the Faculty Evaluator finds that a bound test case or a rubric weight is itself incorrect.
2. The Faculty Evaluator marks the request as a cohort-level defect instead of an individual decision.
3. The system identifies every submission scored against the affected criterion and rubric version.
4. The Faculty Evaluator corrects the rubric or test binding and triggers re-evaluation of the affected submissions.
5. The system revises each affected score, retains the previous values, notifies all affected students and closes the originating request as upheld.